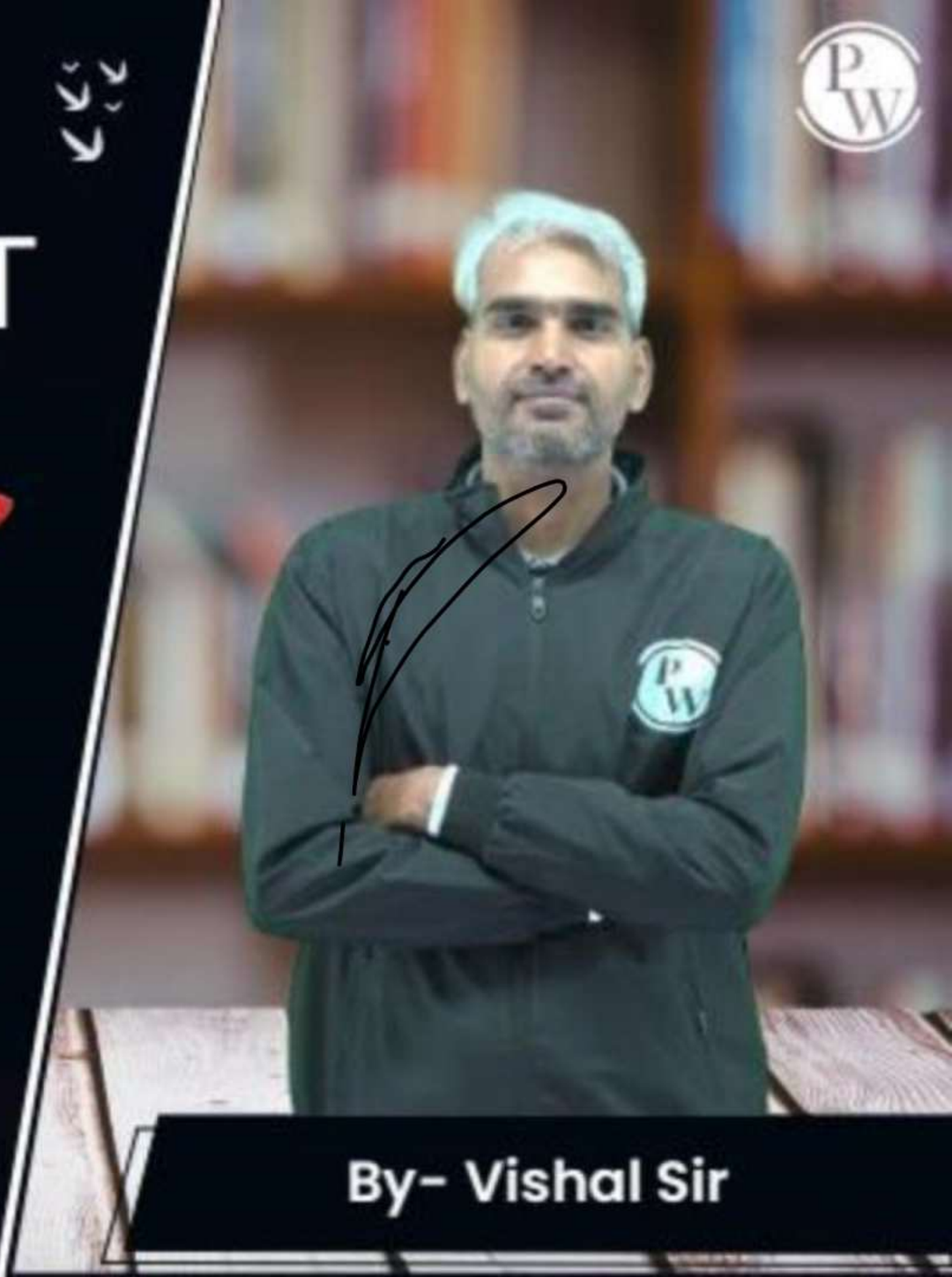
Computer Science & IT



Database management system

Transactions and concurrency control

Lecture No. 08



By- Vishal Sir

Recap of Previous Lecture



Topic Practice questions on conflict serializability

Slide

Topics to be Covered

Topic View Serializable Schedule



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Topic : View serializable schedule

If given schedule is view equivalent to at least one of the serial schedule , then it is called a view serializable schedule.

Converse of the statement is also true



Topic : View equivalent condition

- Consider two schedules S1 and S2
- Schedules S1 and S2 are called view equivalent if the following three conditions hold true for them –
  - Condition-01 : For each data item X, if transaction Ti reads X from the initial database in schedule S1, then in schedule S2 also, Ti must perform the [redacted] read of X from the initial database.

Thumb Rule

- "Initial readers must be same for all the data items".



Topic : View equivalent condition

- Condition-02 :

  If transaction Ti reads the value of data item X updated by the transaction Tj in schedule S1, then in schedule S2 also, transaction Ti must read the value of data item X updated by the transaction Tj.

  Thumb Rule

  - "Write-read sequence must be same.".



Topic : View equivalent condition

- Condition-03 :

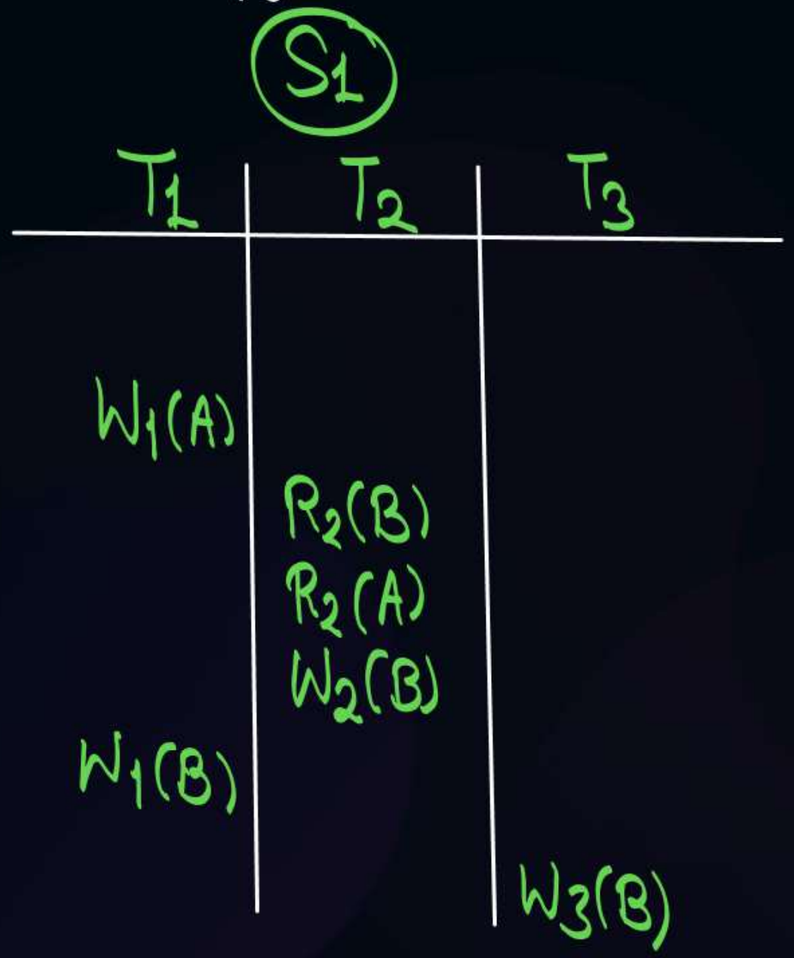
  For each data item X, if X is finally updated by transaction Ti in schedule S1, then in schedule S2 also, X must be finally updated by transaction Ti.

Thumb Rule

- "Final writers must be same for all the data items".

- For two schedules to be view equivalent all three conditions must be satisfied

* Check whether the following schedule is view Serializable Schedule or not? If view serializable schedule, then identify all view equivalent serial schedules.



* Check whether the following schedule is view serializable schedule or not? If view serializable schedule, then identify all view equivalent serial schedules.

S_1

T_1	T_2	T_3
$W_1(A)$		
	$R_2(B)$	
	$R_2(A)$	
	$W_2(B)$	
$W_1(B)$		
		$W_3(B)$

✱1✱ Identify the constraints w.r.t Read from initial db.

Data item	Transactions that reads the dataitem from initial database	Other transactions that update the dataitem	Constraints
A	—	T_1	No Constraint
B	T_2	$\{T_1, T_3\}$	$T_2 \rightarrow \{T_1, T_3\} \equiv T_2 \rightarrow T_1$ & $T_2 \rightarrow T_3$

* Check whethe the following schedule is View Serializable Schedule or not? If View serializable schedule, then identify all view equivalent serial schedules.

S1

T_1	T_2	T_3
$W_1(A)$		
	$R_2(B)$	
	$R_2(A)$	
	$W_2(B)$	
$W_1(B)$		
		$W_3(B)$

2 Identify the Constraints w.r.t. WR sequence:-

Let T_i writes dataitem X in the schedule & T_j reads the value written by T_i and T_k is some other write of dataitem X.

then in equivalent serial schedule T_i must execute before T_j i.e; $T_i \longrightarrow T_j$

and no write of X must be allowed to execute in between T_i & T_j → i.e: $T_k = \phi$ between T_i & T_j

* Check whether the following schedule is View Serializable Schedule or not? If View serializable schedule, then identify all view equivalent serial schedules.

(S1)

T_1	T_2	T_3
$W_1(A)$		
	$R_2(B)$	
	$R_2(A)$	
	$W_2(B)$	
$W_1(B)$		
		$W_3(B)$

(2) Identify Constraints w.r.t Write-Read Sequence :-

Dataitem	Identification of Constraints
A:	T_1 : writes A, and T_2 read the value written by A No other write of A ∴ T_k is already ∅ ∴ Constraint is $T_1 \rightarrow T_2$
B	No Write-Read seq. w.r.t. dataitem B ∴ No Constraint

* Check whether the following schedule is view serializable schedule or not? If view serializable schedule, then identify all view equivalent serial schedules.

S1

T_1	T_2	T_3
$W_1(A)$		
	$R_2(B)$	
	$R_2(A)$	
	$W_2(B)$	
$W_1(B)$		
		$W_3(B)$

3 Identify the Constraints w.r.t. final write

Data item	Transaction with final update	other writers	Constraints
A	T_1	—	No Constraints
B	T_3	$\{T_1, T_2\}$	$\{T_1, T_2\} \rightarrow T_3 \equiv$ $T_1 \rightarrow T_3$, $T_2 \rightarrow T_3$



2 mins Summary

Topic View Serializable Schedule

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THANK - YOU

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